

MD. TANVIR SHEIKH

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building LLM-powered applications: RAG pipelines (both custom-built and LangChain-based), agent orchestration with LangGraph, and tool/function-calling systems via the Groq API. Comfortable across the full stack — embeddings and vector search (ChromaDB), backend APIs (FastAPI), and deployment (Streamlit, Docker) — with a deep-learning foundation in PyTorch. Currently expanding into production evaluation and cloud deployment.

SKILLS

LLM & GenAI : RAG (retrieval-augmented generation), LangChain, LangGraph, agentic tool use, prompt engineering, LLM-as-judge / answer verification, multi-model fallback routing, LLM APIs (Groq), HuggingFace
Vector Search & Embeddings : ChromaDB, HuggingFace sentence embeddings, MMR search, query expansion
ML Frameworks : PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost
Backend & Data : FastAPI, Pydantic, Pandas, NumPy, SQL (MySQL, PostgreSQL, SQLite, Supabase)
Tools & Deployment : Git, Docker, AWS, Streamlit
Languages : Python

PROJECTS

Personal AI Assistant with Corrective RAG — LLM Agent + Retrieval — • [GitHub](#) • [Live Demo](#)

- **Built an agentic AI assistant with LangGraph** orchestrating calculator, web-search, and RAG tools through conditional state-graph routing.
- **Engineered an end-to-end RAG pipeline** with multi-format document ingestion, query expansion, MMR retrieval, deduplication, ChromaDB, and LLM-based relevance filtering.
- **Multi-tenant RAG:** per-session knowledge bases, multi-format document ingestion (PDF/DOCX/TXT/MD), query expansion, MMR search, ChromaDB, and HuggingFace embeddings.
- **Implemented multi-model fallback routing** across Groq-hosted LLMs with automatic rate-limit recovery and Pydantic-validated structured outputs.
- **Developed real-time token streaming** with FastAPI SSE, generation cancellation, multi-thread conversations, and UUID-based session-isolated knowledge bases.
- **Deployed the production application on AWS**, making the AI assistant publicly accessible through a live web interface.

Tech Stack: Python, LangGraph, LangChain, FastAPI, ChromaDB, HuggingFace, Groq, Pydantic, AWS, SSE

Custom RAG Pipeline — Built from Lower-Level Components — • [GitHub](#)

- **Built a RAG system from first principles** instead of high-level LangChain abstractions — custom EmbeddingManager, VectorStoreManager, and RAGRetriever classes — to demonstrate a working understanding of retrieval internals, not just framework calls.
- **Implemented the full ingestion-to-generation pipeline** (PyMuPDF ingestion, recursive chunking, ChromaDB persistent storage, Llama 3.3 70B via Groq) and debugged production-grade retrieval bugs, including a ChromaDB distance-metric inversion and a silent fallback to default embeddings.

Tech Stack: Python, ChromaDB, SentenceTransformers, PyMuPDF, Groq (Llama 3.3 70B)

Attendance by Selfie — CNN-Based Attendance Platform — • [GitHub](#) • [Live Demo](#)

- **Built an end-to-end system** enabling attendance capture via classroom photo or voice recording, with facial recognition (dlib), speaker verification (Resemblyzer), and a Supabase (PostgreSQL) backend.
- **Designed the database schema and resolved data-integrity issues** (foreign-key errors, query bugs) to keep the Supabase backend reliable across role-based student/teacher portals.

Tech Stack: Python, dlib, Resemblyzer, Supabase, Bcrypt, Segno

F1 Pit Stop Prediction — Kaggle Competition — • [Kaggle](#)

- **Built a stacked ensemble** (LightGBM, XGBoost, CatBoost, PyTorch ANN) with GroupKFold cross-validation and domain-specific feature engineering to improve predictive accuracy.

Tech Stack: Python, LightGBM, XGBoost, CatBoost, PyTorch, GroupKFold

EDUCATION

BSc in Computer Science and Engineering — Alhaj Mockbul Hossain University College, Dhaka

April 2022 – Present

Relevant Coursework: Database Management Systems (DBMS), Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Software Development Engineering.